PROJECT DOCUMENTATION

MPI Customer Enquiry Automation

AI-Assisted Lead Management System

Workflow Automation:	n8n
AI Extraction:	Google Gemini API
CRM and Dashboard:	Google Sheets
Version Control:	GitHub

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Prototype and testing documentation

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1 Project Overview

This project automates customer enquiry processing for Moti Packaging Industries (MPI). It uses n8n to orchestrate the workflow, Google Gemini to extract structured information from customer enquiries, and Google Sheets as a lightweight CRM.

The workflow identifies missing information, flags enquiries for human review, prepares customer follow-up drafts, and stores structured lead records.

2 Project Objectives

- Automate the processing of customer enquiries.
- Extract product and order requirements from natural language.
- Identify missing customer and delivery information.
- Check existing records for potential duplicate enquiries.
- Generate customer follow-up drafts.
- Store lead data in Google Sheets.
- Track enquiry metrics through a dashboard.
- Keep customer communication subject to human approval.

3 Technology Stack

Technology	**Purpose**
n8n	Workflow orchestration and automation
Google Gemini API	AI-powered enquiry extraction
Google Sheets	CRM storage and dashboard
JavaScript	Validation and data transformation
GitHub	Version control and documentation

4 Workflow Architecture

The workflow follows this processing sequence:

Manual Trigger → Enquiry Input → Gemini Extract → Validate AI Response

→ Lookup Existing Enquiries → Check Duplicates and Draft Reply → Save Enquiry to Sheet

4.1 Workflow Components

1. **Manual Trigger:** Starts a workflow execution.
2. **Enquiry Input:** Supplies the enquiry ID, source, and raw enquiry text.
3. **Gemini Extract:** Requests structured JSON from the Gemini API.
4. **Validate AI Response:** Parses the AI response and normalizes the extracted fields.
5. **Lookup Existing Enquiries:** Retrieves existing CRM records.
6. **Check Duplicates and Draft Reply:** Checks for potential duplicates, identifies missing fields, and prepares a follow-up draft.
7. **Save Enquiry to Sheet:** Stores the processed record in Google Sheets.

5 Extracted Data Fields

The workflow processes the following information:

- Customer name, company name, and contact details
- Product category and application
- Diameter and height in millimetres
- Order quantity
- Printing and artwork requirements
- Delivery location and required date
- Missing fields and conflicting fields
- Technical concerns and human-review status
- Lead status and follow-up draft

Unknown information should remain missing rather than being invented or guessed.

6 Google Sheets CRM and Dashboard

Google Sheets is used to store the original enquiry alongside the extracted fields and workflow results.

The dashboard summarizes:

- Total enquiries
- Enquiries requiring additional information
- Enquiries requiring human review
- Enquiry errors

A chart provides a visual overview of enquiry metrics. Dashboard formulas depend on the spreadsheet's column layout.

7 Human Review and Safety

Follow-up messages are generated as drafts and are not automatically sent to customers.

Human review is required when information is missing, specifications need verification, or technical concerns need attention. Manufacturing capability, product suitability, pricing, and delivery commitments must be confirmed by the appropriate team.

8 Testing and Results

An end-to-end test was performed using test enquiry ID `MPI-TEST-004`.

8.1 Sample Enquiry

> We need 2000 paper cans, 80 mm diameter and 150 mm height for coffee packaging. Delivery to Delhi. Printed artwork required.

8.2 Observed Extraction Results

Field	Extracted Value
Product category	Paper cans
Diameter	80 mm
Height	150 mm
Quantity	2000
Application	Coffee packaging
Printing requirement	Printed artwork required
Delivery location	Delhi
Lead status	More information required
Human review	Yes

The workflow identified missing customer details and the required delivery date, generated a follow-up draft, and saved the processed record to Google Sheets.

9 Setup and Configuration

1. Import the exported workflow JSON into n8n.
2. Configure Google Sheets credentials and select the appropriate spreadsheet and worksheet.
3. Configure the Gemini API key securely.
4. Verify the Gemini model endpoint and request body.
5. Confirm Google Sheets column mappings.
6. Test with a unique enquiry ID.
7. Verify extracted values, lead status, follow-up draft, and the saved spreadsheet row.

10 Error Handling and Limitations

- Gemini API availability can vary; transient service errors may require retries.
- API errors and malformed responses should be handled and logged safely.
- Duplicate detection depends on configured matching rules.
- The prototype requires broader testing before production use.
- Technical specifications and quotations require human validation.
- Dashboard formulas depend on the spreadsheet column layout.

11 Security

- Never commit API keys, access tokens, or OAuth credentials to GitHub.
- Inspect exported workflow JSON before publishing it.
- Do not publish private customer enquiry data or unrestricted Google Sheets links.
- Use test data in screenshots and public documentation.
- Store credentials securely using n8n credentials or a suitable secrets-management solution.

12 Future Improvements

- Add a dedicated error-handling and notification path.
- Implement a human approval step before customer communication.
- Improve duplicate detection using stable IDs and validated rules.
- Expand testing for missing, conflicting, and malformed information.
- Add lead-source and conversion metrics to the dashboard.
- Introduce production monitoring and controlled retries.

13 Project Status

Prototype demonstrated: Gemini-based extraction, response normalization, Google Sheets CRM integration, follow-up draft generation, and dashboard metrics have been tested with sample enquiries.

Next steps: Expand failure-path testing, finalize approval handling, and validate the workflow under production-like conditions.

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Project: MPI Customer Enquiry Automation
Purpose: AI-assisted enquiry processing and lead management.